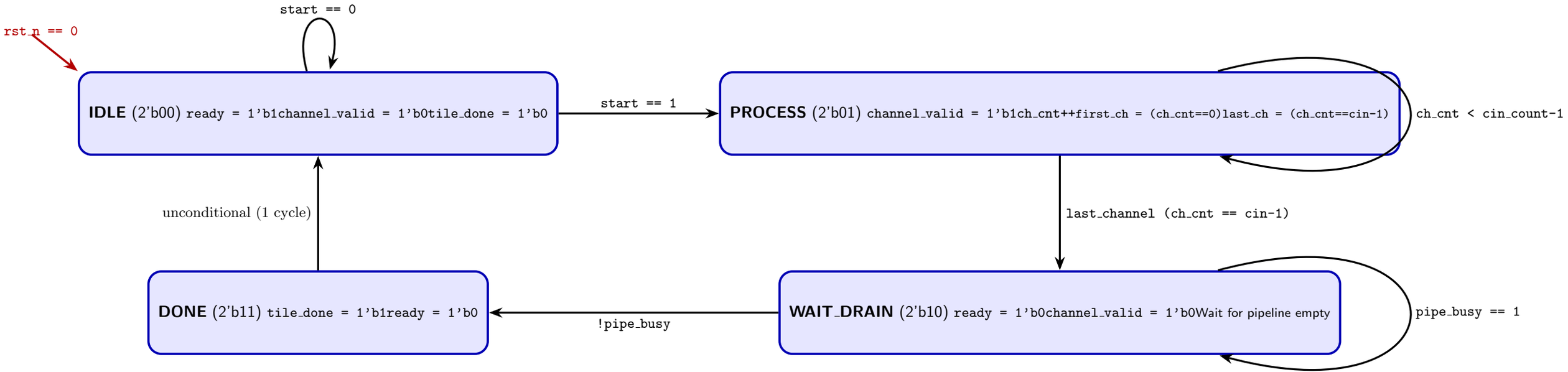

rst_n == 0
start == 0
IDLE (2'b00) ready = 1'b1channel_valid = 1'b0tile_done = 1'b0
start == 1
PROCESS (2'b01) channel_valid = 1'b1ch_cnt++first_ch = (ch_cnt==0)last_ch = (ch_cnt==cin-1)
ch_cnt < cin_count-1
last_channel (ch_cnt == cin-1)
WAIT_DRAIN (2'b10) ready = 1'b0channel_valid = 1'b0Wait for pipeline empty
pipe_busy == 1
!pipe_busy
DONE (2'b11) tile_done = 1'b1ready = 1'b0
unconditional (1 cycle)